



ST BENEDICT'S

SUBJECT	PHYSICS - FINAL EXAM
GRADE	11
EXAMINER	Mr Hilder
MODERATOR	Mrs Govender
DATE	08 November 2024
MARKS	135
DURATION	120 minutes

%

LEARNER NAME:

TEACHER:	GOVENDER	HILDER	NTOMBELA	MCCOY
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PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 20 pages and a Data Sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment.
4. Remember to write your name and surname in the space provided above AND circle your teachers name.
5. Use the data and formulae sheet whenever necessary.
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Answers must be expressed in decimal format, not left as proper fractions.
9. Where appropriate, express answers to TWO decimal places.
10. It is in your own interest to write legibly and to present your work neatly.
11. Blank pages and blank graphs are included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

Question	1	2	3	4	5	6	7	8	9	Total
Mark										
Moderated Mark										
Question Total	14	26	17	17	15	11	11	16	8	135

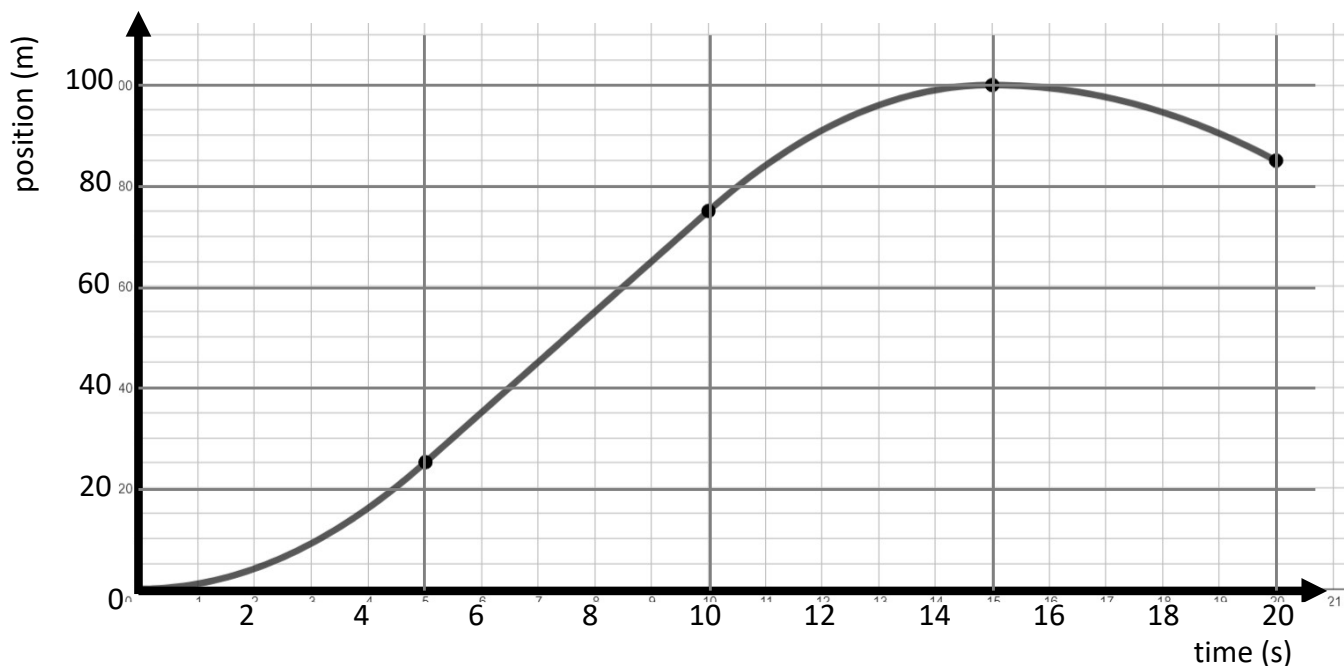
Comment:

Learner's Signature:

Teacher's Signature:

QUESTION 1**14 MARKS**

The graph below shows the position-time graph for a car's motion while travelling east along a straight road. The car experiences four distinct stages of motion:



1.1 Define *acceleration*. (1)

1.2 Calculate the magnitude of the car's acceleration during the first 5 seconds? (3)

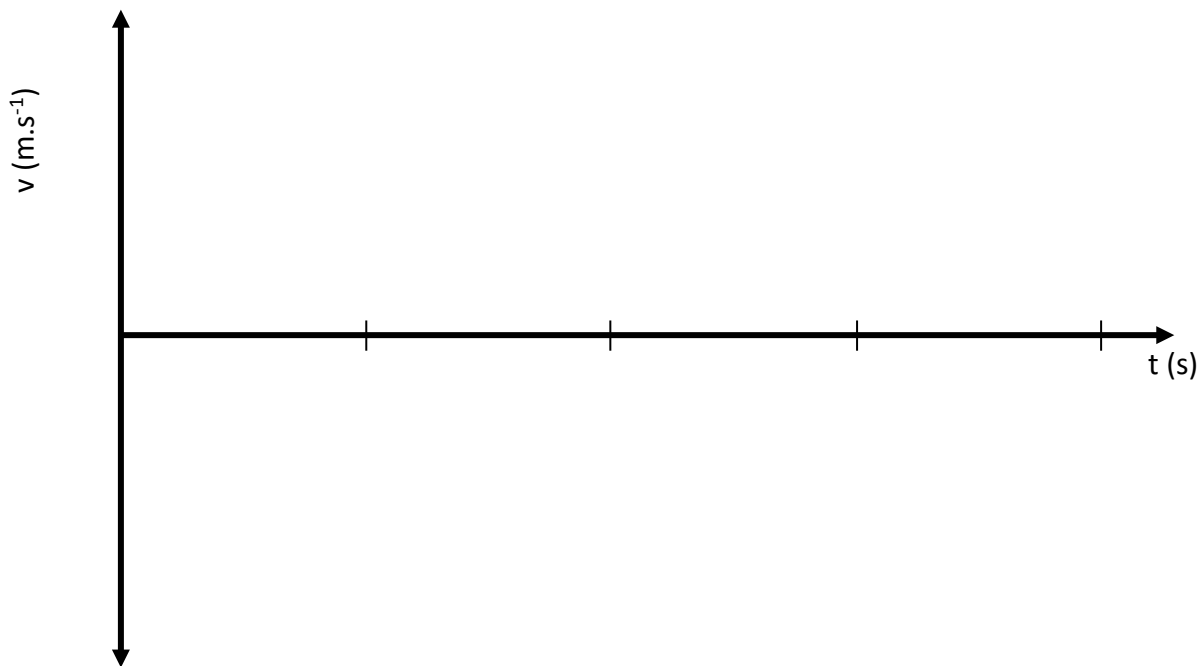
1.3 Determine the car's rate of displacement between 5 and 10 seconds. (2)

- 1.4 Fully describe the car's, acceleration and velocity between 10 and 20 seconds. (4)

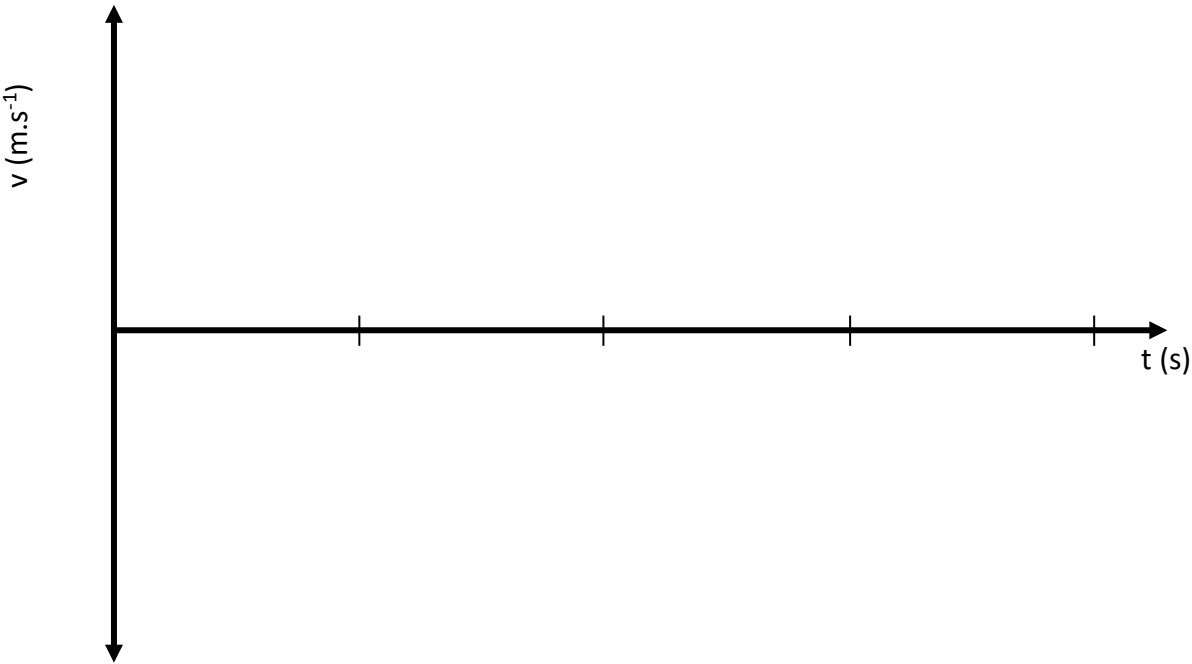
10 – 15 seconds:

15 -20 seconds:

- 1.5 Sketch the corresponding velocity-time graph from the displacement-time graph provided for the entire period from 0 to 20 seconds. Include all known values. (4)

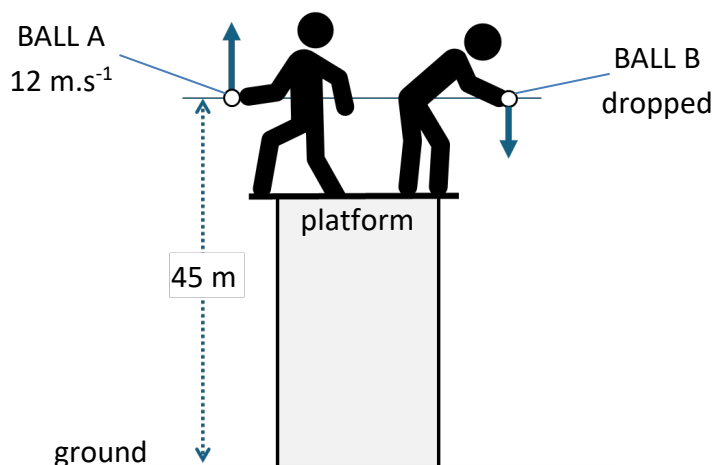


ADDITIAL GRAPH FOR QUESTION 1.5



QUESTION 2**26 MARKS**

A ball, BALL A, is thrown straight up from a height of 45 m, with an initial speed of 12 m.s^{-1} . At the same moment, another ball, BALL B, is simply dropped from the same height, the balls are allowed to bounce once. Air resistance can be ignored.

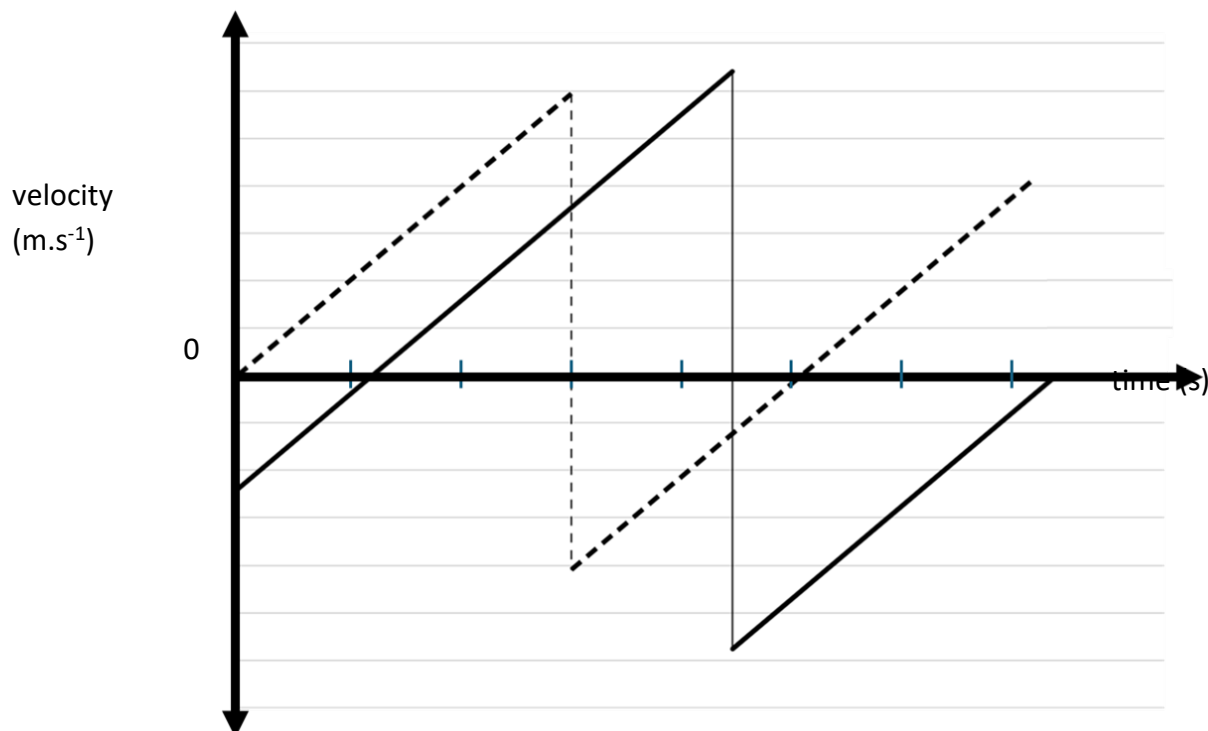


2.1 Calculate the time taken for BALL A to reach its highest point. (2)

2.2 Determine the maximum height above the ground reached by BALL A. (3)

2.3 Calculate the amount time it will take for BALL A to reach the ground. (4)

The velocity-time graph below illustrates the change in velocity of both balls from the moment they leave the platform and bounce. The graph is drawn to scale, but no numerical values are provided.



2.4 For each of the questions below, circle the appropriate answer and explain your answer when required.

2.4.1 Which line represents BALL A? (2)

THE DASHED LINE or THE SOLID LINE

2.4.2 Which ball reaches the ground with a greater velocity? (2)

BALL A or BALL B or THE SAME VELOCITY

2.4.3 Which ball had the greater average velocity when they reached the floor for the first time. (3)

BALL A or BALL B or EQUAL AVERAGE VELOCITY

Briefly explain your answer.

2.4.4 The balls have an equal mass but are made of different materials.

Which ball was made using the less elastic substance?

(2)

BALL A or BALL B.

Briefly explain your answer.

2.5 Indicate (shade or highlight) an area on the velocity time graph on page 6, that shows only the displacement of BALL A, between being thrown and reaching the ground. (2)

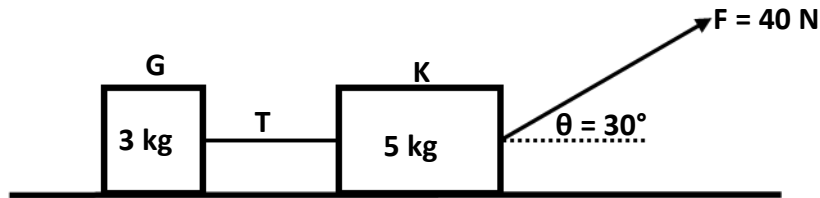
2.6 On its first bounce BALL A bounces high enough for the person to catch it as it reaches its initial height. Prove that the velocity of the ball was approximately $29,7 \text{ m.s}^{-1}$, when it left the ground (3)

2.7 If BALL A was in contact with the ground for 130 milliseconds, determine the average acceleration of the ball during this time. (3)

QUESTION 3**17 MARKS**

Two blocks, G (3 kg) and K (5 kg), are connected by a light, horizontal rope (T) on a rough horizontal surface.

Block K is pulled by a second rope at an angle of $\theta = 30^\circ$ from the horizontal with an applied force $F = 40 \text{ N}$.



The force of friction acting on Block G is 2,5 N, while Block K experience a frictional force of 5,8 N.

3.1 Draw a labelled free body diagram showing all forces acting on Block K. (4)

3.2 Calculate the normal force on Block K. (3)

3.3 Determine is the coefficient of kinetic friction (μ_k) between the surface and Block K. (2)

- 3.4 Describe how the normal force acting on Block K would be affected if the angle (θ) of the applied force (F) was decreased. Explain your answer. (2)

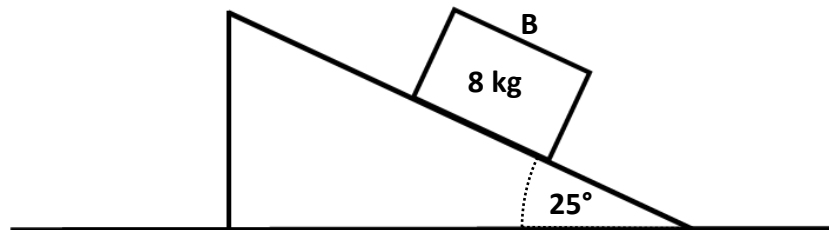
- 3.5 State *Newton's Second Law*. (2)

- 3.6 Calculate the tension in the rope (T) between Blocks G and K. (4)

QUESTION 4**17 MARKS**

Block B of mass 8 kg is placed on a rough inclined plane, which makes an angle of $\theta = 25^\circ$ with the horizontal.

The coefficient of static friction between the block and the inclined plane is $\mu_s = 0,65$.

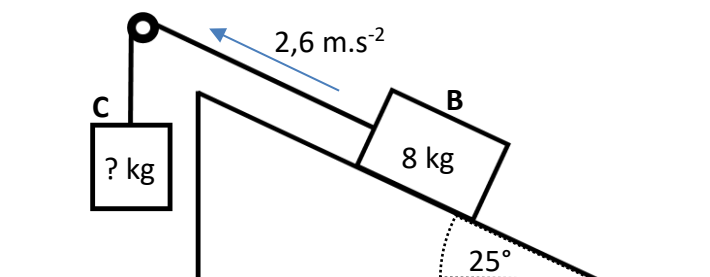


- 4.1 Explain how maximum static friction (F_{fs}^{max}) differs from static friction (F_{fs}). (2)

- 4.2 Determine the magnitude of frictional force acting on Block B. (3)

- 4.3 Draw a vector addition diagram, showing the forces acting on Block B, holding the box stationary. (2)

- 4.4 Block C with an unknown mass is attached to Block B with a rope over a frictionless pulley, the rope acts parallel to the slope.



- 4.4.1 Determine the tension in the rope acting on Block B if the unknown mass of Block C causes both to accelerate at $2,6 \text{ m.s}^{-2}$. The magnitude of kinetic friction (F_{fk}) acting on Block B is $19,75 \text{ N}$ while it is being pulled up the slope. (3)

- 4.4.2 Hence, calculate the mass of Block C. (3)

- 4.5 Circle the appropriate answer and explain your answer when required.

If the angle of the slope (θ) was decreased, how would it affect the:

- 4.5.1 Coefficient of kinetic friction (μ_k) (1)

INCREASE or DECREASE or REMAIN THE SAME.

- 4.5.2 The friction on Block B. (3)

INCREASE or DECREASE or REMAIN THE SAME.

Briefly explain your answer.

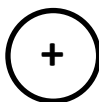
QUESTION 5**15 MARKS**

A student investigates how the electric field (E) at a fixed distance from a charged sphere depends on the sphere's charge (Q). The student charges a sphere positively and measures E at the same distance from the centre each time, varying only the charge Q on the sphere. The results are recorded in the table below.

Q (nC)	E (kN·C ⁻¹)
2	6
5	19
7	25
9	34
12	41
14	52

5.1 Define *electric field at a point*. (1)

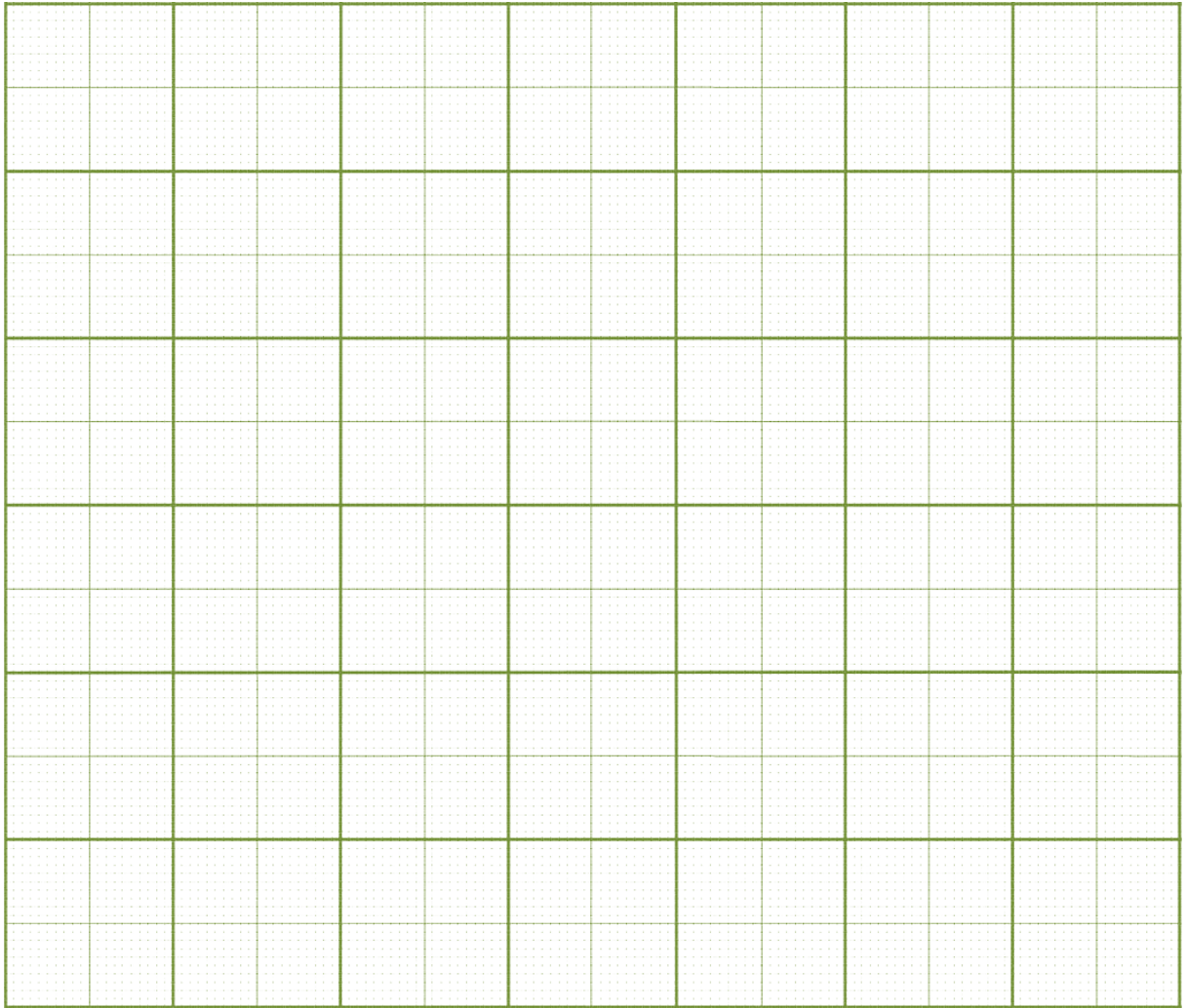
5.2 Draw the electric field lines around a positively charged sphere (2)



5.3 Name the independent variable in the experiment. (2)

5.4 Plot a graph of electric field (on the y-axis) vs charge (on the x-axis) on the axis provided on **page 13**. No heading is required. (5)

5.5 Calculate the gradient of the graph. Indicate the values you used for this calculation on your graph. Include the unit of the gradient. (4)

GRAPH FOR QUESTION 5.4

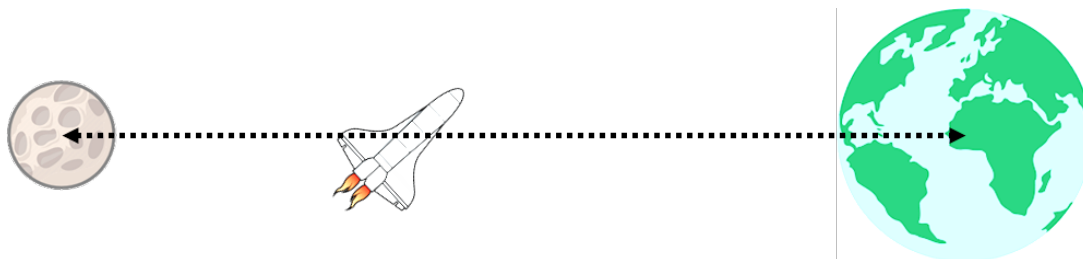
- 5.6 Use your answer from Question 5.5, your knowledge of electric fields and that the equation describes a straight-line to determine the distance from the charge at which the student made all the measurements of electric field. Express your answer to three decimal places. (2)

QUESTION 6**11 MARKS**

A spaceship of mass $m_s = 1,500 \text{ kg}$ is positioned between Earth and the Moon along the line joining their centres.

The mass of Earth (m_E) is $5,97 \times 10^{24} \text{ kg}$ and the mass of the Moon (m_m) is $7,35 \times 10^{22} \text{ kg}$.

The centre-to-centre distance (d) between Earth and the Moon is. $3,84 \times 10^8 \text{ m}$.



- 6.1 Define *Newtons Universal Law of Gravitation*. (2)

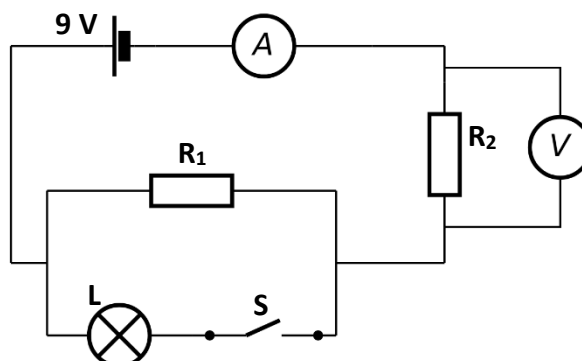
- 6.2 Calculate the gravitational field strength exerted by Earth at the position of the spaceship, if the spaceship is $3 \times 10^8 \text{ m}$ from Earth's centre. (3)

- 6.3 Determine the net gravitational field strength at this point due to both Earth and the Moon, including the direction of the resultant field. (4)

- 6.4 What is the weight of the spacecraft in this position in space? (2)

QUESTION 7**11 MARKS**

A circuit consists of a battery with an emf of 9 V connected to a resistor network, as shown below. Ignore internal resistance of the battery and connecting wires.



When switch S is closed, the bulb (L) dissipates 5 Watts, the ammeter reads 0,82 A. R_1 has a resistance of $5\ \Omega$ and R_2 a resistance of $7\ \Omega$.

Switch S is closed.

7.1 Explain what is meant by the term *resistance*. (2)

7.2 What is reading on the voltmeter (V)? (2)

7.3 Determine the resistance of Bulb L under these conditions. (4)

Switch S is open.

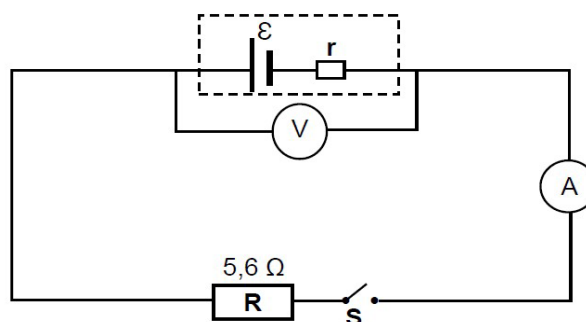
7.4 Calculate the new current in the battery. (2)

7.5 How will the voltage reading of the voltmeter change?

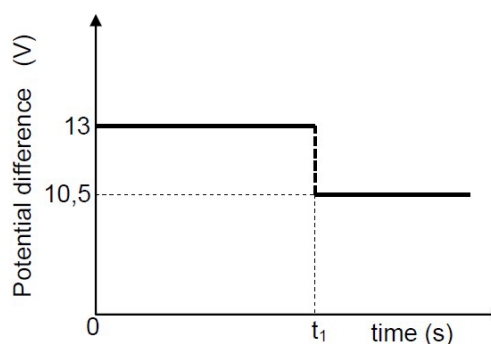
Circle: INCREASE, DECREASE OR REMAIN THE SAME? (1)

QUESTION 8**16 MARKS**

In the circuit diagram below, resistor **R**, with a resistance of $5,6\ \Omega$, is connected with a switch, an ammeter and a high-resistance voltmeter, to a battery with an unknown internal resistance, **r**. The resistance of the connecting wires and the ammeter may be ignored.



The graph below shows the potential difference across the terminals of the battery as a function of time. At time t_1 , switch **S** is closed.



8.1 Define *emf*. (2)

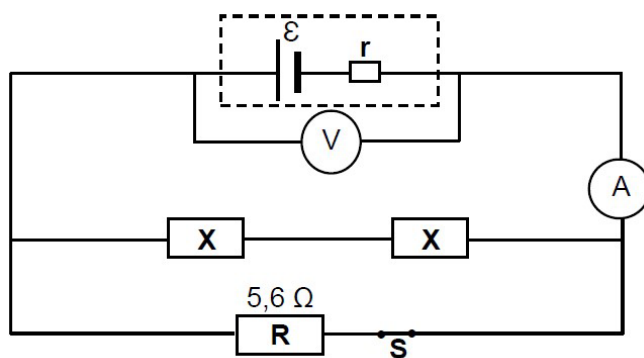
8.2 Write down the value of the emf of the battery. (1)

8.3 When switch **S** is CLOSED, calculate the:

8.3.1 current through resistor **R**. (2)

8.3.2 internal resistance, **r**, of the battery. (3)

- 8.4 Two IDENTICAL resistors, each with resistance X , are now connected in the same circuit with switch S closed, as shown below.



The ammeter reading now increases to 3 A.

- 8.4.1 How would the voltmeter reading change?

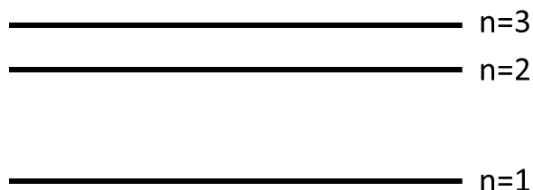
Circle: INCREASES, DECREASES or REMAINS THE SAME. (1)

- 8.4.2 Justify your answer in Question 8.3.1 referring to two relevant formulae. (3)

- 8.4.3 Calculate resistance X . (4)

QUESTION 9**8 MARKS**

An energy level diagram, for a particular element, is drawn to scale in the diagram below. The only positions that the outer electron can occupy are the energy levels $n=1$, $n=2$, and $n=3$.



9.1 Which transition would release light with the highest frequency? (2)

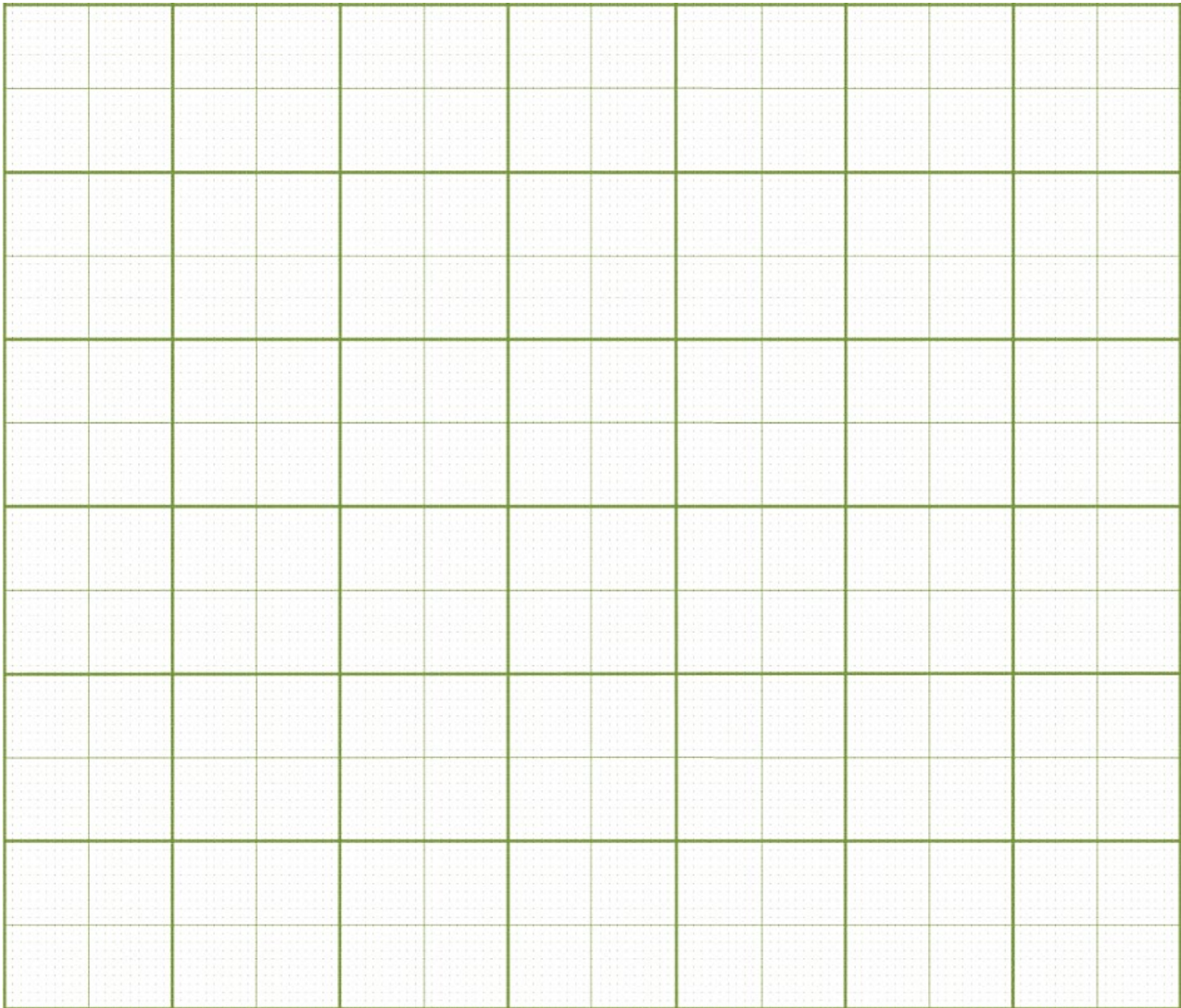
For this energy-level diagram, the longest wavelength possible is 618 nm.

9.2 On the diagram, **above**, show the transition for this wavelength. (2)

9.3 Calculate the energy in eV that corresponds to a wavelength of 618 nm. (4)

END OF ASSESSMENT

ADDITIONAL GRAPH PAPER



ADDITIONAL SPACE

[illegible]

ACKNOWLEDGEMENTS:

- IEB NSC NOVEMBER - PAPER I – 2016
- HILTON COLLEGE – PRELIMINARY EXAM -PAPER I - 2024